Bihar Engineering University, Patna

B.Tech 4th Semester Examination, 2024

Course: B.Tech
Code: 101403

Subject: Engineering Geology

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin
*(ii) There are **NINE** questions in this paper*
*(iii) Attempt **FIVE** questions in all*
(iv) Question No. 1 is compulsory

Q.1 *Choose the correct option / write the answer of the following (Any seven):* [2 x 7 = 14]

(a) The property of a mineral by virtue of which it can be cut with a knife is
(i) Parting
(ii) Sectile
(iii) Malleable
(iv) Ductile

(b) When the mineral occurs in flattened or square form, it is called
(i) Tabular
(ii) Elongated.
(iii) Batroidal
(iv) Box

(c) Kyanite shows which form?
(i) Columnar
(ii) Elongated
(iii) Bladed
(iv) Tabular

(d) The arrangement made near top of dam for passage of excess water from the reservoir.
(i) Sluice way
(ii) Spillway
(iii) Gallery
(iv) Abutments

(e) Chromite shows which type of structure?
(i) Granular
(ii) Globular
(iii) Reniform
(iv) Mammillary

(f) A temporary dam constructed to exclude water from a specific area.
(i) Debris dam
(ii) Diversion dam
(iii) Coffer dam
(iv) Storage dam

(g) What occurs due to fall of big rock blocks due to release of stresses during tunneling?
(i) Rock fall
(ii) Rock burst
(iii) Blockage
(iv) Water rush

(h) Mica is
(i) Flexible
(ii) Rigid
(iii) Flexible and elastic
(iv) Elastic

(i) Granite is which type of rock
(i) Igneous Rock
(ii) Sedimentary Rock
(iii) Metamorphic Rock
(iv) All of these

(j) Hardness of human finger nail is :
(i) 1.5
(ii) 2
(iii) 3
(iv) None of these

Q.2 (a) What is Rock forming mineral? Discus the process of formation of mineral in nature. [7]

(b) Give a Detailed account on the physical properties of feldspar. [7]

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Q.3 (a) Write the engineering properties and uses of [7]
a) Granite
b) Dolerite
c) Basalt
d) Sandstone.

(b) Distinguish between Igneous and sedimentary rocks. [7]

Q.4 (a) Enumerate the mitigation measures for Slope instability. [7]
(b) What are the factors that influence slope instability? [7]

Q.5 (a) Illustrate the various types of folds with neat sketches. [9]
(b) Write a note on seismic zone in India. [5]

Q.6 (a) Illustrate the various types of fault with neat sketches. [7]
(b) Critically discuss the concept of plate tectonics and its role in generation of Earthquake. [7]

Q.7 (a) Discuss briefly on Rock Bolting , Rock Anchoring and Retaining Wall [7]
(b) Discuss the creation of reservoir. Draw a sketch to show a dam with reservoir and label it. [7]

Q.8 (a) What is Dam and its Utility? Draw the sectional view of a dam. [7]
(b) Write an essay on geological condition required for design and construction of dam. [7]

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Q.9 Answer *any two* of the following [7 x 2=14]
(a) Explain the geological causes for the failure of the Dam.
(b) Explain the suitability of different Rocks in Dam construction.
(c) Enumerate with appropriate figures on the types. causes of landslides and their mitigation measures.

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